

INTERNAL DOCUMENTATION

ERP System Business Flow

End-to-end documentation of all operational flows —
procurement, sales fulfillment, last-mile delivery,
workforce management, and finance.

Version	Date	Modules
1.0	June 2026	AUTH · HRIS · INVENTORY · WMS · TMS · FINANCE

System Modules

The ERP system is composed of six modules. **WMS is the operational hub** — it sits at the intersection of inventory, logistics, and finance, orchestrating the core fulfillment pipeline.

AUTH
Authentication & session management. JWT-based login, token refresh, and user profile. Foundation for all other modules.

HRIS
Human Resources — employee records, shift scheduling, attendance tracking, and login audit logs.

INVENTORY
Product catalog, warehouse bin locations, and real-time stock levels. Central ledger for all physical goods.

WMS
Warehouse operations — inbound receiving & putaway, outbound picking, QC checking, dispatch, and invoice generation.

TMS
Transportation — route planning, GPS tracking, delivery confirmation (POD), and driver fuel logging.

FINANCE
Accounts Payable (AP) and Accounts Receivable (AR) — invoices, payments, aging summaries, and branch management.

Cross-Module Dependencies

WMS → INVENTORY
Receiving goods adds stock; confirming picks deducts stock from bins.

WMS → TMS
Dispatching a sales order triggers creation of a TMS delivery route.

WMS → FINANCE
Fulfilled SOs generate AR invoices; POs feed AP invoices.

HRIS → WMS / TMS
Employee records used when assigning pick tasks, putaway, check tasks, and driver routes.

Procurement-to-Pay

Covers the inbound goods journey from supplier purchase order through to payment. Managed by Operations Managers, Warehouse teams, and Finance.

Inbound Receiving Workflow

WMS + INVENTORY + FINANCE

- 1 Create Purchase Order** WMS
Operations Manager creates a PO with supplier and product lines. Status: *pending*.
- 2 Receive Goods at Warehouse** WMS
Warehouse Supervisor or Operator records received quantities against the PO. A Goods Receiving Note (GRN) is created. Stock is added to INVENTORY at a staging bin.
- 3 Generate Putaway Tasks** WMS
Supervisor generates putaway tasks from the GRN. Tasks are assigned to a Warehouse Operator.
- 4 Execute Putaway** WMS · INVENTORY
Operator scans or manually assigns each item to a bin location. On confirmation, stock is moved from staging to the target bin. INVENTORY levels updated.
- 5 Create & Pay Supplier Invoice** FINANCE
Finance Officer creates an AP invoice linked to the PO. Payment is recorded when processed, updating the AP aging report.

Roles Involved

Step	Role	Action
Create PO	operations_manager	Initiates purchase from supplier
Receive Goods	wh_supervisor · wh_operator	Logs received quantities against PO
Generate Putaway	wh_supervisor	Creates and assigns putaway tasks
Execute Putaway	wh_operator	Moves goods to bin locations
AP Invoice / Payment	finance_officer	Records payable and payment

Order-to-Cash (Sales Fulfillment)

The outbound order pipeline from sales order creation through picking, quality check, dispatch, and payment collection.

Outbound Fulfillment Workflow

WMS + TMS + FINANCE

- 1 Create Sales Order** WMS
Operations Manager creates an SO specifying customer, delivery date, and ordered products. Status: *pending*.
- 2 Generate Pick Tasks** WMS · INVENTORY
Supervisor generates pick tasks from the SO. Each task maps a product to its bin location. Tasks assigned to Warehouse Operators.
- 3 Execute Picking** WMS · INVENTORY
Operator picks items from bins and confirms each line. Reserved stock is deducted from INVENTORY. SO status moves to *packed*.
- 4 Quality Check** WMS
Checker reviews all lines of the picked order against expected quantities.

FAIL — returns to picking

PASS — ready to dispatch
- 5 Dispatch & Invoice** WMS → TMS · FINANCE
Supervisor dispatches the SO — this triggers a TMS delivery route. WMS generates a Sales Invoice, which appears in Finance as an AR receivable.
- 6 Collect Payment** FINANCE
Finance Officer records customer payment against the AR invoice. Aging report updated.

Roles Involved

Step	Role	Action
Create SO	operations_manager	Initiates customer order
Generate Picks	wh_supervisor	Auto-generates tasks per SO line
Execute Picks	wh_operator	Physical picking from bins
Quality Check	checker	Verify packed quantities
Dispatch	wh_supervisor · dispatcher	Release order to logistics
AR Payment	finance_officer	Records customer payment

Last-Mile Delivery

The physical delivery execution from route creation through GPS tracking, proof of delivery, and fuel logging.

Delivery Execution Workflow TMS

- 1 Create Delivery Route** TMS
Dispatcher creates a route with a vehicle, driver, and ordered delivery stops. Each stop is linked to a dispatched Sales Order.
- 2 Driver Starts Route** TMS
Driver opens the mobile app, views today's route, and taps "Start Route." Route status changes to *in_progress*.
- 3 Real-Time GPS Tracking** TMS
Mobile app sends GPS pings while driving. Operations team can monitor all active vehicles on a live map dashboard.
- 4 Confirm Delivery (POD)** TMS
Driver confirms each stop with optional notes and photo (Proof of Delivery). Stop status: *delivered*.
- 5 Log Fuel** TMS
After completing the route, driver selects the completed route, enters liters used and distance driven. System computes fuel efficiency (km/L) automatically.

Roles Involved

Step	Role	Action
Create Route	<code>dispatcher</code> · <code>operations_manager</code>	Plan stops and assign driver/vehicle
Start Route	<code>driver</code>	Begin delivery on mobile app
GPS Tracking	<code>driver</code> (sends) · <code>dispatcher</code> (views)	Real-time vehicle monitoring
Confirm Delivery	<code>driver</code>	POD with photo per stop
Fuel Log	<code>driver</code>	Post-route fuel and distance entry

Workforce Management

Employee lifecycle from onboarding through daily shift and attendance management, with full login audit visibility.

HRIS Workflow **HRIS**

- 1 Create Employee Account** **HRIS**
HR Manager creates a user record with employee code, full name, department, and assigned role. The role controls what the employee can access across all modules.
- 2 Assign Shifts** **HRIS**
HR creates shift records (start time, end time, shift type) for employees. Shifts can be morning, afternoon, or night.
- 3 Daily Attendance** **HRIS**
Employees clock in and out. Attendance records are linked to assigned shifts. Status is tracked (present, absent, late).
- 4 Login Audit** **HRIS**
Every login is recorded (timestamp, device type, IP address). HR and Operations Managers can view login history per employee with device icons for quick identification.

Role Hierarchy & System-Wide Access

Role	Modules	Key Responsibilities
<code>system_admin</code>	All modules	Full access to every operation and record
<code>operations_manager</code>	HRIS, Inventory, WMS, TMS, Finance	Create orders, manage inventory, oversee all operations
<code>hr_manager</code> <code>hr_staff</code>	HRIS	Employee records, shift scheduling, attendance
<code>wh_supervisor</code>	Inventory, WMS	Generate pick/putaway/check tasks, dispatch orders
<code>wh_operator</code>	WMS, Inventory	Execute picks, receive goods, putaway
<code>checker</code>	WMS	Quality control of packed orders before dispatch
<code>dispatcher</code>	WMS, TMS	Create delivery routes, monitor GPS, manage vehicles
<code>driver</code>	TMS	Start routes, confirm deliveries, log fuel
<code>finance_officer</code>	Finance	AP/AR invoice management and payment recording